

Predicting the morphological changes of braided sand-bed rivers with deep learning

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Abstract—Magherini [1] introduced *JamUNet*, a U-Net-based model trained on annual satellite-derived binary images to predict braided river planform evolution one year ahead. Building on this baseline, we investigate three directions to improve data-driven morphodynamic prediction: (i) a ConvLSTM architecture to explicitly model temporal ordering, (ii) dataset expansion through image splitting and the inclusion of additional sand-bed braided river segments, and (iii) a ternary representation (water/sand/vegetation) to enrich semantic information. While the gains brought by explicit temporal recurrence through the ConvLSTM architecture remained moderate, experiments on the new three classes datasets show that both deep models consistently outperform a no-change baseline. Data augmentation meanwhile proved inconclusive and constrained by data quality and computational limits.

I. INTRODUCTION

Magherini’s thesis [1] demonstrated that deep learning can provide a practical alternative to physics-based models for predicting the planform evolution of braided sand-bed rivers. Using pre-classified annual satellite imagery of the Brahmaputra–Jamuna River, *JamUNet* predicts the river mask at year $t+1$ from the four preceding years and improves over a persistence (no-change) benchmark.

In this project, we extend Magherini’s framework along three complementary axes. First, we introduce a ConvLSTM model to explicitly capture spatio-temporal dependencies instead of treating multi-year unordered inputs. Second, we explore data extension strategies, including spatial splitting of reaches and the addition of other large sand-bed braided river segments to increase sample diversity. Third, we construct a ternary dataset (water, sand, vegetation) to better represent geomorphological features and test whether the three class dataset improves binary planform prediction.

II. METHODS

A. ConvLSTM model

1) *Structure*: The ConvLSTM model extends classical LSTMs by replacing fully connected operations with convolutions, enabling joint modeling of spatial and temporal dependencies. In this work, the network takes as input a sequence of four annual raster maps and predicts the next river planform through pixel-wise segmentation. The architecture consists of stacked ConvLSTM layers that preserve spatial structure while learning temporal evolution through recurrent

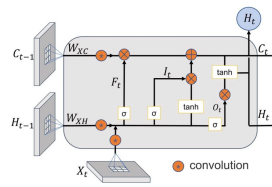


Fig. 1. Single ConvLSTM cell.

hidden states (Figure 1). Following Shi et al. [2], the final hidden state is projected to the output space via a 1×1 convolution. Two model variants are implemented for binary and ternary datasets.

2) *Hyperparameter tuning and training strategy*: Due to computational constraints, extensive hyperparameter optimization was not feasible. A full grid search with cross-validation would have required prohibitive training times, so a simplified one-factor-at-a-time (OFAT) strategy was adopted. Model selection was guided by the F1-score, with a weighted F1-score used for the ternary task. Hyperparameters with limited computational cost (e.g., learning rate, class thresholds) were explored more extensively, whereas architectural parameters (number of layers, hidden channels, kernel size) strongly affected training time. Although larger models improved F1 performance, they quickly became impractical. Figure 2 shows the evolution of the best F1-score across tested hyperparameters, while per-epoch computational time analyses are reported in the Appendix A.

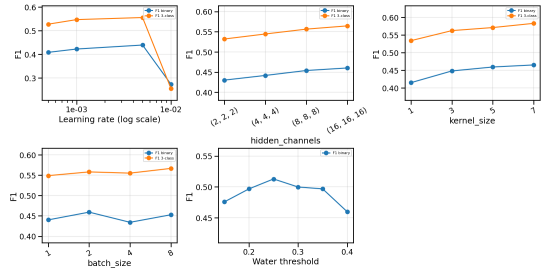


Fig. 2. Hyperparameter tuning results: Best F1-score as a function of learning rate, model capacity, kernel size, batch size, and water threshold.

The final configuration was therefore selected as a trade-off between performance and feasibility. All models were trained using the Adam optimizer, chosen for its robustness to noisy gradients and suitability for segmentation tasks. A StepLR learning-rate scheduler was employed to stabilize convergence during training. Binary classification relied on a binary cross-entropy loss, while ternary classification used a categorical cross-entropy loss.

B. Ternary satellite-based classification of the Brahmaputra–Jamuna River

This section presents the pipeline developed to construct a ternary (*water, sand, vegetation*) satellite-derived dataset for the Brahmaputra–Jamuna River. The approach extends the binary water/non-water representation introduced by Magherini

[1] while remaining compatible with JamUNet and ConvLSTM architectures. Dedicated diagnostics were used to assess class balance and interannual transitions.

1) *Study area and data extraction:* The spatial domain and river reaches follow the definition of A. Magherini. Corresponding shapefiles were imported into Google Earth Engine (GEE) to extract surface reflectance imagery from the Landsat collections LT05, LE07, LC08, and LC09 (Collection 2, Tier 1, Level 2) [3], [4], [5], [6]. Only the spectral bands required to compute NDVI and MNDWI were retained (Appendix C).

Data extraction was restricted to four months per year corresponding to the low-flow season, during which channel morphology is most clearly observable. Images were exported at 30 m spatial resolution, requiring a subsequent resampling step to ensure compatibility with downstream preprocessing.

2) *Cloud contamination:* Only limited cloud masking was applied. Although more advanced cloud-filtering approaches exist [7], they are largely sensor-specific and could not be consistently applied across the multi-sensor Landsat archive. Residual cloud contamination and image artifacts therefore remain a known limitation of the dataset.

3) *Classification and preprocessing:* A threshold-based NDVI/MNDWI classification was initially tested but discarded due to poor temporal robustness. Instead, an unsupervised k -means clustering approach was adopted, grouping NDVI and MNDWI values into three classes corresponding to water, sand, and vegetation, while excluding no-data pixels. This method relies on relative spectral separation and is more robust to interannual variability, but cannot explicitly distinguish clouds from bright surfaces such as sand.

To mitigate cloud-induced misclassification, one visually optimal image per year and reach was manually selected from the four candidate months. The resulting ternary maps were then processed using the preprocessing pipeline developed by A. Magherini, including cropping, normalization, imputation of missing values, and formatting into fixed-size raster tensors. Apart from the additional resampling step, the pipeline was left unchanged to preserve comparability with the binary dataset.

4) *Dataset diagnostics:* Class composition statistics and year-over-year transition matrices were computed to characterize dataset properties and interannual dynamics (Figure 11, Appendix F). Transition rates were normalized by class availability in the initial year, yielding availability-adjusted transition percentages.

C. Data expansion

1) *Image size reduction:* Image sizes were reduced from $60 \text{ km} \times 30 \text{ km}$ (1000×500 pixels) to $30 \text{ km} \times 30 \text{ km}$ (500×500 pixels; Figure 3) by splitting each rectangular image in two. The resulting tiles were preprocessed following the original pipeline. This procedure was applied exclusively to the

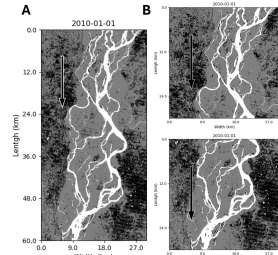


Fig. 3. Splitting of January 1st, 2010 input image. **A** Original rectangle image. **B** Image split in two 500×500 tiles.

Jamuna reaches, effectively doubling the amount of training data.

2) *Inclusion of additional rivers:* To increase the dataset size, the option of considering other portions of the world's largest sand-bed braided rivers was considered. This approach is supported by the assumption Bristow and Best (1993)[8] stating that the the bed grain size does significantly influence the large-scale braiding processes and could possibly enable a generalization of the model to all braided rivers. Among various possibilities, a portion further north of the Brahmaputra-Jamuna river was first selected. The reach layout was designed on QGIS[9], uploaded on GEE and exported with the JRC Monthly History GSWD[10] collection following the same workflow as Magherini's. The satellite view of the reaches is displayed in Appendix C. Figure 4 displays an overview of

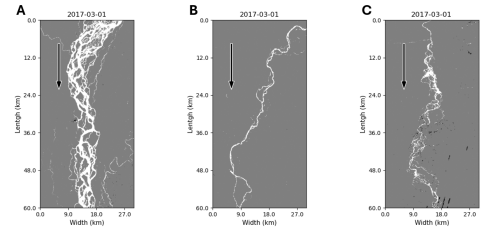


Fig. 4. 2017.03.01 Jamuna, Gange and Xigaze segments after pre-processing. **A** Brahmaputra-Jamuna, training reach 26. **B** Gange, training reach 5. **C** Xigaze portion of Jamuna, training reach 3.

training reaches of the Jamuna, Gange and Xigaze segments in March 2017 after pre-processing.

III. RESULTS

A. Experiment: Binary dataset (Brahmaputra–Jamuna)

We first evaluate the performance of the proposed ConvLSTM model on the same binary dataset used by Magherini [1], consisting of 732 annual satellite images of the Brahmaputra–Jamuna River. Each pixel is classified as either water or non-water, and the task is to predict the river planform at year $t+1$ from the four previous annual observations. The ConvLSTM is compared against the original JamUNet architecture and a no-change baseline, which directly uses the last input image as prediction. Results metrics of the two models and the benchmark are shown in Appendix A.

B. Experiment: Binary dataset (Extended Dataset)

After pre-processing, addition of the Xigazê and Gange segments to the original dataset allowed to dispose of combined training, testing and validation sets of 1108, 56 and 56 samples, respectively, which is approximately twice as big as the initial datasets. Table I displays results for the training of JamUNet on this augmented data for 10 epochs (batch size 1, over 600 min) and training on ConvLSTM with lightened hyperparameters (batch size 8, downsample factor 4, kernel size 3, padding 1) for 30 epochs.

C. Experiment: Binary data with reduced image size

For the image size reduction experiments, models trained on 500×500 pixel inputs showed consistently low and non-improving F1 scores (0.5), indicating insufficient learning

TABLE I
PERFORMANCE ON THE EXTENDED DATASET FOR UNET AND CONVLSTM

Metric	UNet (10 epochs)	ConvLSTM (30 epochs)
BCE Loss	3.268×10^{-1}	0.3166
Accuracy	0.8810	0.8228
Precision	0.7060	0.4733
Recall	0.5120	0.6751
F1 score	0.5190	0.5457
CSI	0.3930	0.4213

of large-scale spatial structures. These configurations were therefore not evaluated further, and no quantitative results are reported.

D. Ternary dataset generation and quality assessment

The ternary classification pipeline produced annual maps of the Brahmaputra–Jamuna River distinguishing water, sand, and vegetation. The resulting class imbalance, aggregated over all reaches and years, is summarized in Table II.

TABLE II
MEAN CLASS PROPORTIONS IN THE TERNARY DATASET

Class	Mean proportion (%)
Water	14.13
Sand	24.87
Vegetation	61.00

Figure 5 illustrates the sensitivity of the classification to satellite image quality on the validation reach. Clean images yield spatially coherent and physically consistent ternary maps, whereas images affected by cloud contamination or reconstruction artifacts produce degraded classifications with reduced spatial consistency. These observations justify the manual selection of one visually optimal image per year. One key

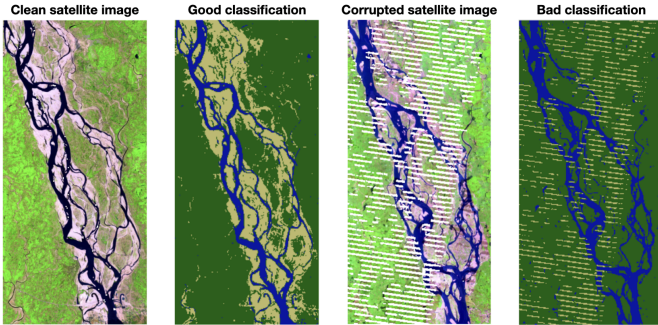


Fig. 5. Impact of satellite image quality on ternary classification on the validation reach. (Left) Clean satellite image and corresponding accurate ternary classification. (Right) Corrupted satellite image affected by reconstruction artifacts and resulting degraded classification.

improvement compared to the JRC dataset was the ability to retrieve quality data between 1993 and 1999 by using multiple Landsat datasets.

Year-over-year transition statistics were computed to characterize morphological dynamics. The mean availability-adjusted transitions between classes are shown in Figure 11 (Appendix F). A larger sample of classified images can be visualized in the Python notebook `classification_pipeline.ipynb`.

E. Experiment: Ternary dataset and binary evaluation

Using the newly introduced ternary dataset (water, sand, vegetation), we conduct a second set of experiments designed to assess the impact of richer semantic inputs on binary river planform prediction. Two training protocols are considered. In the first protocol, models are trained to perform ternary segmentation, and their outputs are subsequently mapped to binary predictions by merging sand and vegetation into the non-water class. In the second protocol, the same dataset is converted to binary labels prior to training, and models directly predict binary outputs. For each protocol, three models are evaluated: ConvLSTM, JamUNet, and the no-change baseline. This setup enables a fair comparison across models and input representations while keeping the evaluation objective strictly binary. The corresponding results are shown in Table IV. A

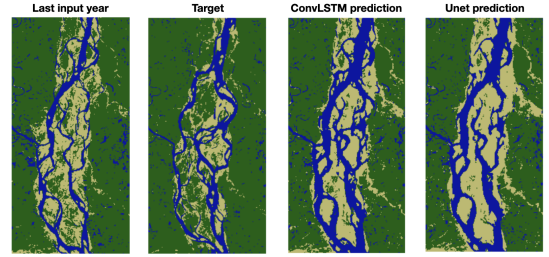


Fig. 6. Example visualization of three class predictions of ConvLSTM and Unet with target image

TABLE III
F1-SCORES FOR BINARY RIVER PLANFORM PREDICTION USING THE TERNARY DATASET UNDER TWO TRAINING PROTOCOLS.

Model	3-class training	Binary training
ConvLSTM	0.5718	0.5426
JamUNet	0.5768	0.5375
No change	0.4740	0.4740

IV. DISCUSSION

A. Comparison of ConvLSTM, JamUNet and No change on Binary Dataset

Although the ConvLSTM was designed to explicitly capture temporal dependencies, the observed improvement over JamUNet remains modest. The ConvLSTM achieves only a marginal gain in F1-score and CSI, while overall performance levels and BCE loss are nearly identical for both models. This suggests that, in the current setting, temporal recurrence provides limited additional information beyond what is already implicitly captured by the stacked multi-year inputs of JamUNet. One likely explanation is that river planform evolution at annual resolution is relatively slow and smooth, making short-term temporal dependencies weak and reducing the benefit of recurrent memory. In addition, computational constraints severely limited model capacity and hyperparameter exploration; larger ConvLSTM configurations consistently improved performance but were impractical to train. Nevertheless, both learning-based models clearly outperform the no-change baseline across all skill-based metrics, confirming that they effectively learn meaningful spatio-temporal patterns beyond simple persistence.

B. Data augmentation

For both the UNet and ConvLSTM models, the data augmentation strategy based on image size reduction did not outperform training on the original dataset, within the constraint of a reduced number of training epochs. Preliminary results suggest that the reduced spatial context of the split images (30×30 km instead of 60×30 km) is less effective at capturing the large-scale spatial structures governing braided river morphodynamics. Consequently, key patterns of channel migration and erosion–deposition dynamics may be only partially represented, leading to degraded predictive performance.

The second augmentation strategy, based on extending the training dataset with additional river systems, also yielded lower performance than the baseline dataset under the same computational constraints. Two factors likely contributed to this outcome. First, the added rivers (the Ganges and the Xigazê portion of the Jamuna) have significantly narrower riverbeds than the original Brahmaputra–Jamuna segment, increasing class imbalance and introducing different, thinner spatial structures (Figure 4).

Second, cloud-induced perturbations and reconstruction noise were observed in the Xigazê test set (Figure 10, Appendix C). These artifacts, inherent to the JRC dataset rather than the preprocessing workflow, likely degrade the quality of the supervision signal and hinder generalization.

Although these results do not challenge the general effectiveness of dataset extension as a means of improving model performance, they highlight the importance of data homogeneity and quality. In this case, the inclusion of morphologically heterogeneous and noisy data limited the benefits of augmentation. Overall, these findings emphasize that data augmentation in geophysical learning tasks must be carefully designed, with close attention to data consistency, noise characteristics, and physical relevance.

C. Discussion on ternary classification

The proposed ternary classification extends the binary river planform representation previously used for the Brahmaputra–Jamuna River. A key improvement is the ability to retrieve usable classified imagery for the period 1993–1999, which was a major limitation of the JRC-based dataset. Consequently, all reaches contain at least 30 well-classified years out of 35, enabling the construction of long temporal sequences suitable for learning-based models.

Under favorable imaging conditions, the unsupervised k -means classification reliably separates water, sand, and vegetation, as confirmed by visual inspection. Although class imbalance remains, the ternary formulation is more informative than the binary one, as it explicitly distinguishes mobile sand bars from more stable vegetated areas. This added semantic structure is supported by the transition statistics, which show strong mutual exchange between water and sand, while vegetation exhibits weaker and more symmetric transitions. These patterns are consistent with active channel dynamics and suggest that the ternary labels capture physically meaningful variability.

Several limitations nevertheless affect the usefulness of the dataset for machine learning. In some years, all candidate images were corrupted by clouds or reconstruction artifacts, preventing reliable classification. More generally, cloud contamination and image noise introduce label uncertainty, particularly in vegetated areas, which likely reduces achievable predictive performance. Attempts to explicitly cluster clouds using k -means were unsuccessful, as the algorithm lacks mechanisms to identify cloud-contaminated pixels and led to excessive data loss.

These limitations indicate that the primary bottleneck of the ternary approach is data quality rather than the clustering strategy itself. Improved cloud handling and noise mitigation are therefore expected to have a larger impact on model performance than further classifier tuning. In particular, incorporating a k -nearest neighbors (KNN)–based interpolation or masking strategy prior to clustering could stabilize the spectral space and reduce misclassifications. While not implemented here, such approaches represent a promising direction for future work.

Overall, the ternary classification can be considered successful in terms of data coverage and physical consistency, but its full potential for improving learning-based river morphology prediction is currently constrained by satellite data quality rather than model capacity.

D. Ternary dataset and binary evaluation

The results of the last experiment indicate that incorporating richer semantic information through ternary supervision has a limited but noticeable impact on binary river planform prediction. When models are trained on the full three-class task and subsequently mapped to binary outputs, both ConvLSTM and JamUNet achieve higher F1-scores than when trained directly on binary labels, suggesting that explicit modeling of sand and vegetation helps the networks learn more informative intermediate representations. However, the performance gap between ConvLSTM and JamUNet remains small and inconsistent across protocols, with JamUNet slightly outperforming ConvLSTM under 3-class training, and ConvLSTM marginally improving under binary training. This further supports the conclusion that, under current computational constraints and temporal resolution, explicit temporal modeling provides only limited additional benefit. Nevertheless, both learning-based models substantially outperform the no-change baseline, confirming that they effectively exploit the added semantic richness of the ternary dataset to improve binary planform prediction beyond simple persistence.

V. CONCLUSION

This study explored three complementary extensions of the JamUNet framework for data-driven prediction of braided river morphodynamics: explicit temporal modeling with ConvLSTM, dataset expansion, and a ternary semantic representation of river surfaces. While ConvLSTM and ternary supervision consistently outperformed a no-change baseline,

their improvements over the original JamUNet remained modest, highlighting the strong baseline performance of spatial architectures and the limited influence of short-term temporal recurrence at annual resolution. Data augmentation strategies proved sensitive to data quality and morphological heterogeneity, emphasizing the importance of physically consistent and noise-controlled datasets. Overall, this work confirms the relevance of deep learning for river planform prediction while identifying data quality, temporal resolution, and computational constraints as key limiting factors, pointing toward future improvements through enhanced satellite preprocessing and longer, cleaner temporal records.

VI. ETHICAL CONSIDERATIONS

A first ethical risk associated with data augmentation through the inclusion of additional river systems lies in the implicit selection bias introduced during dataset construction. The choice of which rivers are incorporated directly affects the domains over which the model generalizes reliably, potentially privileging regions that are represented while disadvantaging others. The stakeholders most impacted are populations and local decision-makers living near braided rivers that were not included in the training data, as model predictions for these systems may be less accurate or unreliable. In applied contexts such as infrastructure planning or risk assessment, this could lead to unequal access to dependable predictive tools. The severity of this risk is moderate to high when model outputs inform real-world decisions, and its likelihood is moderate, as users may assume broad generalizability from multi-river training. At the project level, it was not possible to ensure representative coverage of all braided river systems. If further research was conducted, this risk should be addressed by explicitly limiting the claimed domain of validity of the model, transparently documenting dataset selection, and cautioning against extrapolation to unrepresented river systems without additional validation.

A second ethical consideration concerns the environmental cost associated with increased model complexity. Although the ConvLSTM architecture yields slightly improved performance compared to JamUNet, this gain comes at the cost of substantially higher computational requirements, particularly during hyperparameter tuning and training. The recurrent nature of ConvLSTM significantly increases training time and energy consumption, especially under constrained hardware conditions where extensive experimentation becomes costly and inefficient. From an ethical perspective, this raises questions about the proportionality between marginal performance improvements and their environmental footprint. Given that JamUNet achieves comparable predictive performance with a simpler architecture and lower computational demand, favoring ConvLSTM in practical deployments may not always be justified. This trade-off was explicitly considered during model evaluation, and results were reported transparently to enable informed decision-making. In contexts where sustainability, accessibility, or limited computational resources are critical factors, retaining a simpler model such as JamUNet may represent a more responsible and environmentally conscious choice, without significantly compromising predictive quality.

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APPENDIX A CONVLSTM DESIGN

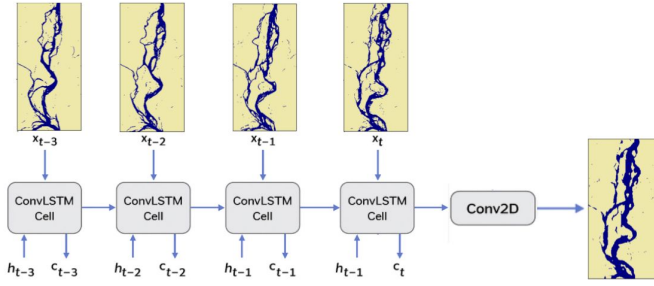


Fig. 7. Overview of the ConvLSTM design and architecture.

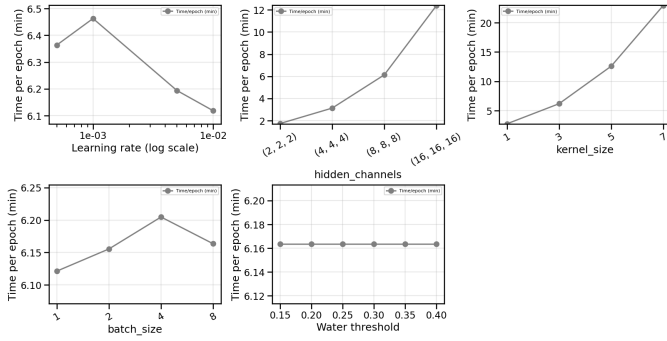


Fig. 8. Per epoch computational time as a function of learning rate, model capacity, kernel size, batch size and water threshold

TABLE IV
AVERAGE PERFORMANCE METRICS ON TEST SET

Metric	JamUnet	ConvLSTM	No change
BCE loss	0.151	0.151	6.082
Accuracy	0.923	0.925	0.920
Precision	0.627	0.634	0.641
Recall	0.816	0.809	0.683
F1 score	0.707	0.709	0.659
CSI score	0.547	0.550	0.493

APPENDIX B REPOSITORY NAVIGATION

The new data folder is organized with the following structure:

```
data/
|-- 500x500_datasets/
|-- combined_datasets/
|-- flow/
|-- qgis/
|-- satellite/
|-- satellite_gange/
|-- satellite_rakaia/
|-- satellite_xigaze/
|-- __init__.py
|-- bathymetry_1992.csv
```

|-- README.md

The `satellite_gange`, `satellite_rakaia` and `satellite_xigaze` folders contain data for the Ganges, the upper portion of the Jamuna and the Rakaia, respectively, following the same organization as `satellite`. The Rakaia dataset was abandoned because the river is much thinner compared to the other rivers and for the same reach size, the data is unusable. The `combined_datasets` folder contains the concatenated training, validation and testing datasets from all three rivers (Jamuna, Gange, Xigaze).

Generation and pre-processing of the 500x500 data was performed using the notebook `preproc_500x500.ipynb`. Datasets generation and UNet training were subsequently performed using the notebook `Unet3D_train_val_spatial_dataset_500x500.ipynb`, calling adapted dataset generation functions from the script `dataset_generation_split.py`. In this script, the dataset generation functions were solely modified to handle the split files (images and averages) in the format `*_top` and `*_bottom` corresponding to upper and lower halves of the original data, respectively. Besides, the split data was systematically stored in sub-folders of `datasatellite` in the form:

```
satellite/
|-- averages                # 1000x500
|-- averages_split          # 500x500
|-- preprocessed            # 1000x500
|-- preprocessed_split      # 500x500
|-- dataset_month3          # 1000x500
|-- dataset_month3_split    # 500x500
... 
```

Similarly, the split March data after pre-processing is stored in `dataset_month3_split` and the split training, testing and validation datasets are stored in the folder `500x500_datasets`.

The training of the UNet on the extended dataset with concatenated training, validation and testing reaches was done in the notebook `Unet3D_train_val_spatial_extended_dataset.ipynb`

APPENDIX C SPECTRAL INDICES USED FOR TERNARY CLASSIFICATION

Vegetation and water surfaces were characterized using spectral indices derived from multispectral satellite imagery. These indices are computed from specific optical wavelength ranges commonly available in Landsat sensors. The red and green bands correspond to the visible spectrum (approximately 0.63–0.69 μm and 0.52–0.60 μm , respectively), while the near-infrared (NIR, approximately 0.77–0.90 μm) and shortwave infrared (SWIR, approximately 1.55–1.75 μm) bands lie beyond the visible range and are sensitive to vegetation structure and surface moisture.

The Normalized Difference Vegetation Index (NDVI) exploits the contrast between near-infrared and red reflectance and is defined as

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}}$$

High NDVI values indicate vegetated areas due to strong NIR reflectance and red absorption by chlorophyll, whereas values close to zero are typically associated with bare soil or sand, and negative values with water.

Open water surfaces were enhanced using the Modified Normalized Difference Water Index (MNDWI), defined as

$$\text{MNDWI} = \frac{\rho_{\text{Green}} - \rho_{\text{SWIR}}}{\rho_{\text{Green}} + \rho_{\text{SWIR}}}$$

Water bodies exhibit relatively high reflectance in the green band and strong absorption in the SWIR band, leading to positive MNDWI values, while vegetation and exposed sediment generally yield negative values. These indices were jointly used to discriminate water, sand, and vegetation in the ternary classification [7].

APPENDIX D DATA QUALITY FOR OTHER RIVERS

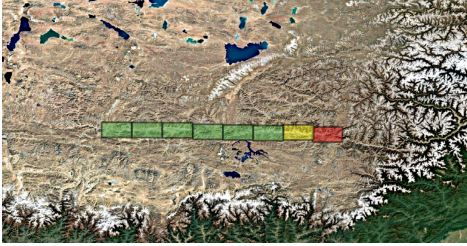


Fig. 9. QGIS Satellite view of the North portion of the Brahmaputra-Jamuna river in Tibet, next to Xigazê district. Reaches destined to training are in green, while yellow and red indicate validation and testing portions, respectively.

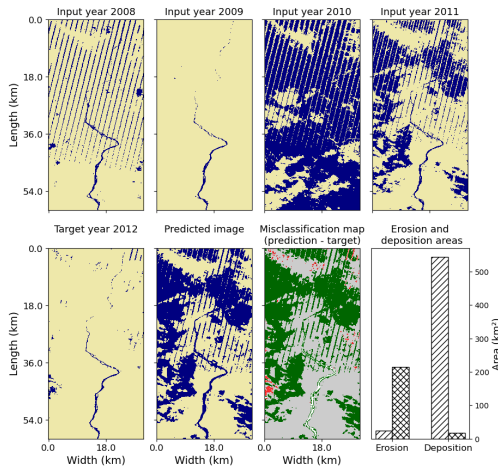


Fig. 10. Prediction on Xigaze testing reach, input years 2008-2011, target year 2012

Figure 9 shows the satellite view of the layout of the training, validation and testing reaches for the Xigazê segment.

Figure 10 clearly highlights the presence of noise on the binarized satellite images from the JRC Monthly History GSWD collection[10]. On one hand, the noisy areas in the form of irregular patches picked up as water (e.g. Input year 2008) are typical of cloud noise; on the other hand, the thin aligned stripes (e.g. Input year 2010) are often due to overlapping satellite image reconstruction. It is safe to assume that in their original form, cloudy images perturb the model more than they benefit it. Extensive pre-processing would be required to make this data usable.

APPENDIX E MEAN CLASS TRANSITION STATISTICS

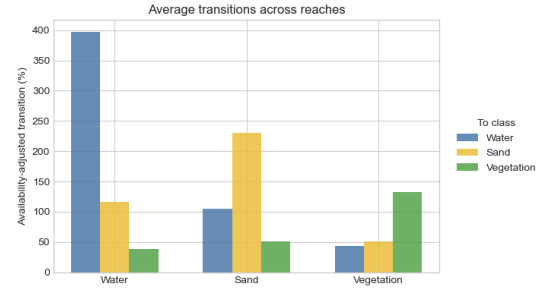


Fig. 11. Mean availability-adjusted year-over-year transitions between classes, averaged across all reaches. Bars indicate the percentage of pixels transitioning from a given source class (x-axis) to each target class, normalized by the availability of the source class in the initial year.

Figure 11 shows the mean availability-adjusted year-over-year transitions between classes, averaged across all reaches. The transition rates are normalized by the availability of each source class in the initial year, allowing for a direct comparison despite strong class imbalance.

An interesting pattern emerges from these transition statistics. Water and sand exhibit a stronger mutual exchange, with transitions between these two classes occurring more frequently than transitions from either class to vegetation. In contrast, vegetation displays comparable transition rates toward water and sand, suggesting a weaker coupling with either class. This asymmetry indicates a closer dynamical relationship between water and sand, consistent with their shared association with active channel processes, while vegetation appears to behave as a more independent and stabilizing component of the river corridor.